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# A Simple Coverage Model for Missing Variant Calls

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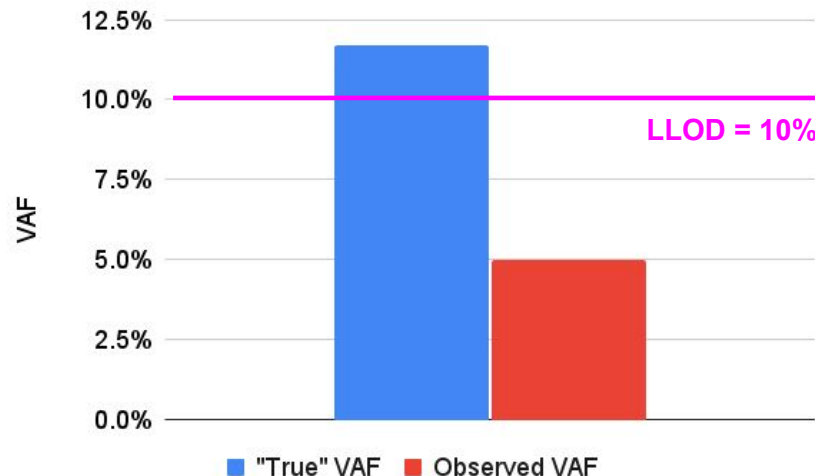
# A Story of Rage Coding



*Rage coding:* When rage at a situation motivates rapid coding to resolve it.

# The Situation

- CAPNGSST proficiency testing
- We are given a sample to sequence and identify variants
  - Variant in *MET* c.3028+1G>T p.?
  - OICR observed VAF was **5%**
  - But CAP reported VAF was **11.7%**
  - “True” VAF from digital PCR above our lower limit of detection (**LLOD**) of **10%**
- Flagged during the ACD inspection (November 2025)



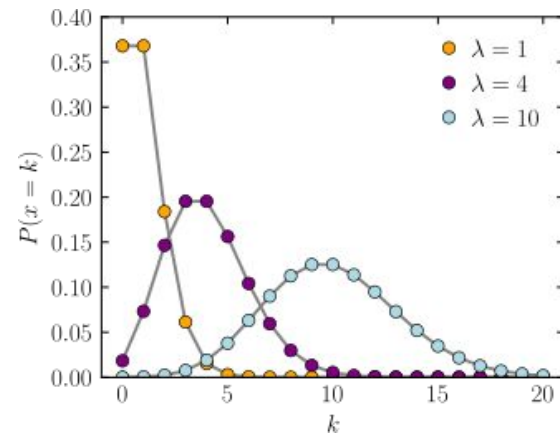
# The Reaction

- **Intuition:** The missed call is not surprising, given random variation in depth
- But how to prove it?



# The Poisson Distribution

- After French mathematician Siméon Denis Poisson (1781-1840)
- Probability of a given number of events occurring in a fixed interval
- The events must occur:
  - With a known constant mean rate
  - Independently of previous events
- Good for lots of things
  - Radioactive decay
  - Short read sequencing!
- For sequencing the “interval” is a location on the genome
  - Reads not truly independent, but close enough



# An Equation for Coverage

The Poisson distribution can be used to model any discrete occurrence given an average number of occurrences.

The probability function is the following:

$$P(Y=y) = (C^y e^{-C})/y!$$

- y is the number of times a base is read
- C stands for coverage

– *Illumina Technical Note, “Estimating Sequencing Coverage”*

# The Question

- The equation gives us the probability of *any particular* coverage value
- We are interested in the outcome of two coverage values, variant and non-variant
- What is the probability of *all outcomes* which fall below our LLOD?
  - Imagine coverage has a *perfectly smooth* depth of 80X
  - At a true VAF of 11.7%, we expect 9 variant reads and 71 non-variant reads
  - But coverage is not perfectly smooth!

# A Simple Model for Variant Coverage

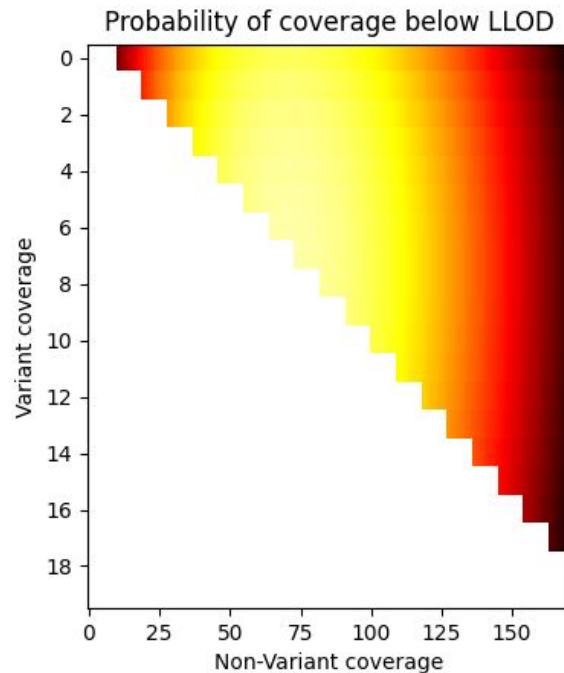
- We model coverage as two Poisson distributions:
  - Y: Non-variant reads, mean depth of coverage  $A = 71$
  - Z: Variant reads, mean depth of coverage  $B = 9$
- The outcome which interests is a pair of depths  $(y, z)$
- Assume the values of Y and Z are independent
- Then we can multiply the two probabilities:

$$\begin{aligned}\Pr(Y=y, Z=z) &= \Pr(Y=y)\Pr(Z=z) \\ &= (A^y e^{-A} B^z e^{-B}) / y! z!\end{aligned}$$



# Likelihood of coverage below LLOD

- We want to sum the probabilities of the area in which the call is **not** made
- Coloured area is below LLOD = 10%



# Coverage and LLOD

Finally, we take the sum over values of Y and Z with VAF below LLOD:

$$\Pr(\text{VAF} < \text{LLOD}) = \sum_{(y,z: z/z+y < \text{LLOD})} \Pr(Y=y)\Pr(Z=z)$$

In our example:

- Variant mean depth = 9
- Non-variant mean depth = 71
- $\Pr(\text{VAF} < \text{LLOD}) = 0.37$

**Further explanation is not needed for this variant**



# Detecting a “True” VAF Just Above LLOD Is Hard!

- True VAF = 11.7%
- LLOD = 10%



# Future work

- Apply the model to other proficiency test results (one just came in)
- Check how well the *aggregate* of our variant calls fits the simple model
- Code (35 lines of Python) is in:  
[https://github.com/oicr-gsi/simple\\_coverage\\_model](https://github.com/oicr-gsi/simple_coverage_model)



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